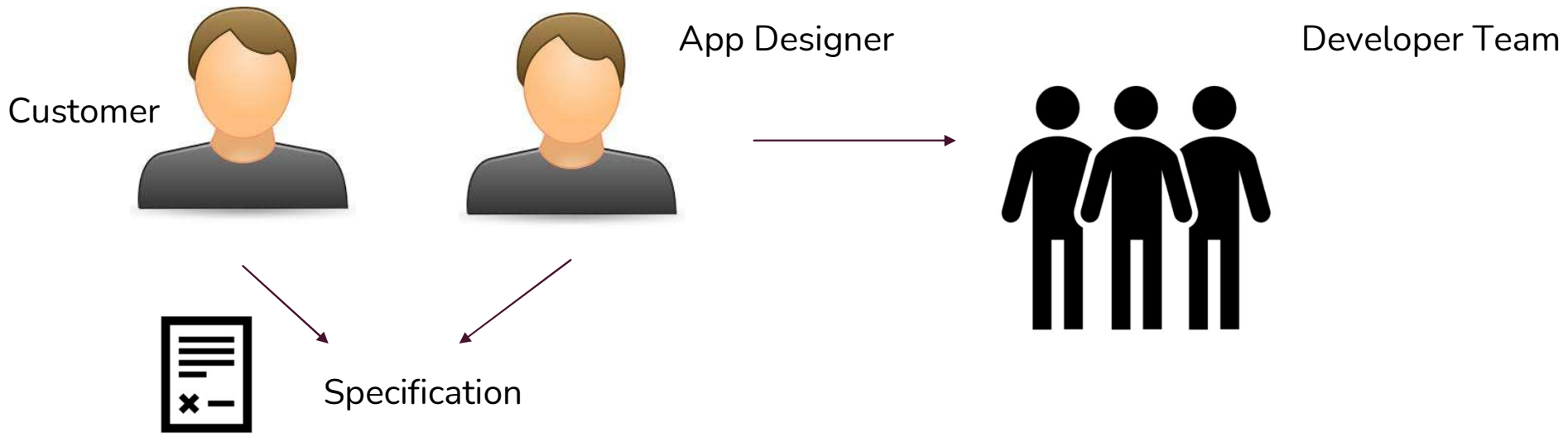
PRESENTATION LAYER

SELECTED RESOURCES

- https://developer.android.com/develop/ui/compose/
- https://developer.android.com/courses/jetpack-compose/course
- https://github.com/android/compose-samples/tree/main/Jetchat
- https://developer.android.com/topic/architecture/recommendations

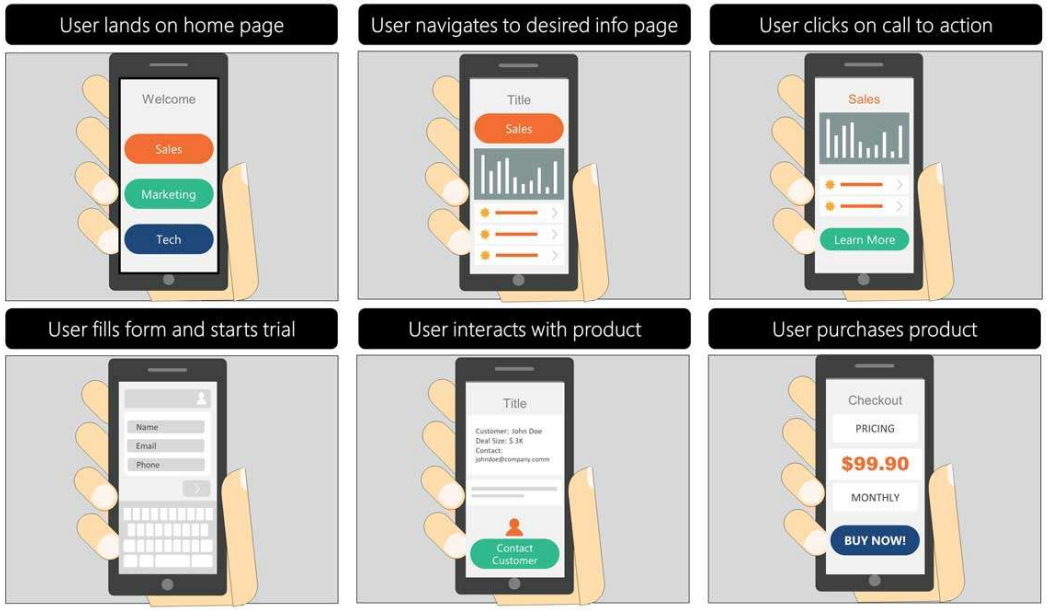
PRELUDE: APP SPECIFICATION

- When developing an app, the first step is to understand **what** the user expects to see and **how** they will interact with app.
- The design of UI (user interface) isn't then just aesthetics: it's logic, communication, (and psychology).
- The process starts with a specification of the requirements, which is agreed between a customer and the designer



PREDULE: STORYBOARD

- A storyboard is a visual representation of basic screen flow that shows key user interface components and basic user interaction to achieve a user goal.
- It useful for brainstorming on requirements.



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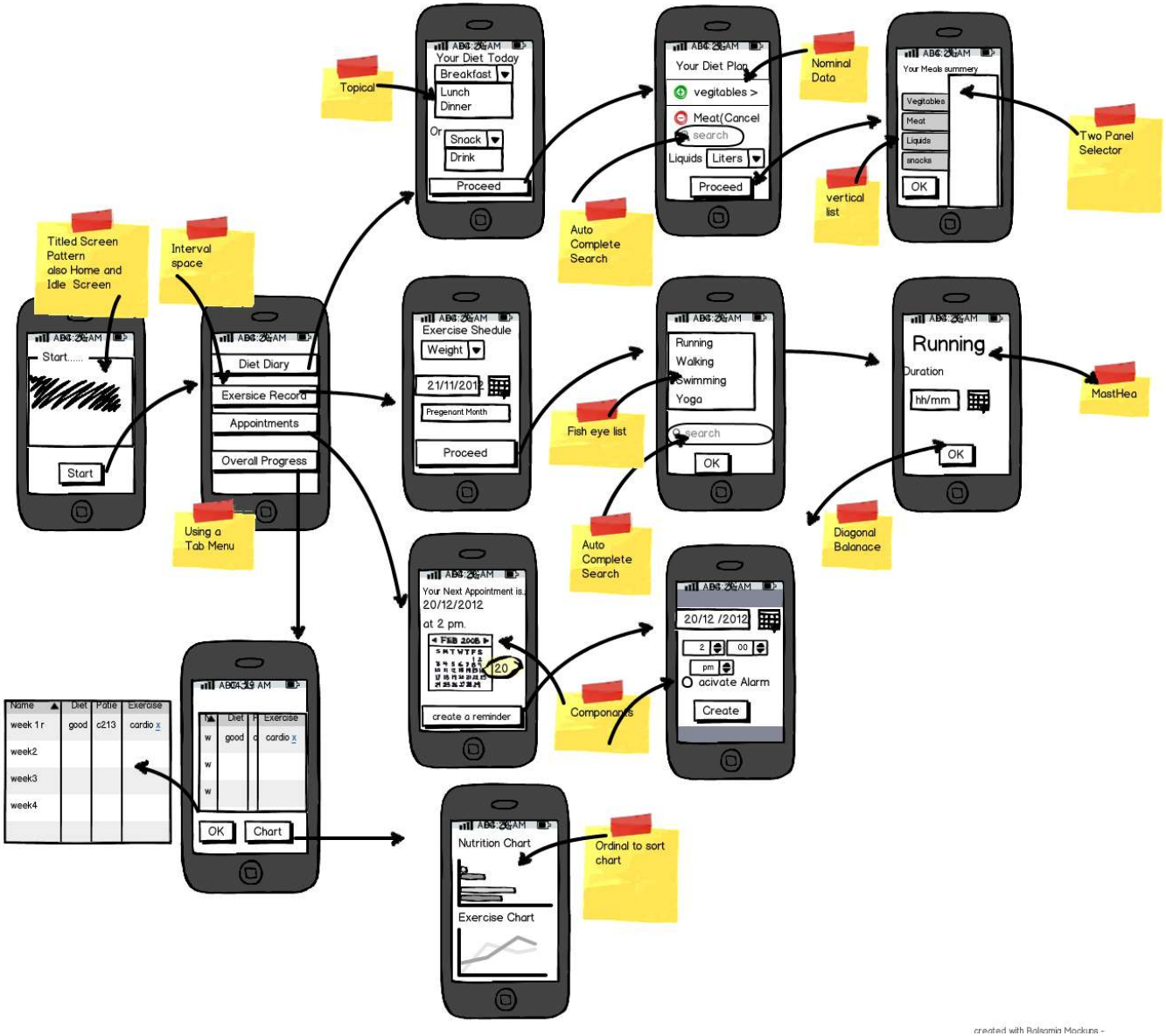
EXAMPLE


Titled Screen Pattern also Home and Idle Screen
Interval space
Topical
Nominal Data
Two Panel Selector
Auto Complete Search
vertical list
Fish eye list
Auto Complete Search
Diagonal Balanace
Using a Tab Menu
Componants
Ordinal to sort chart
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Overall Progress
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Lunch
Dinner
Or
Snack
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snacks
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Exercise Shedule
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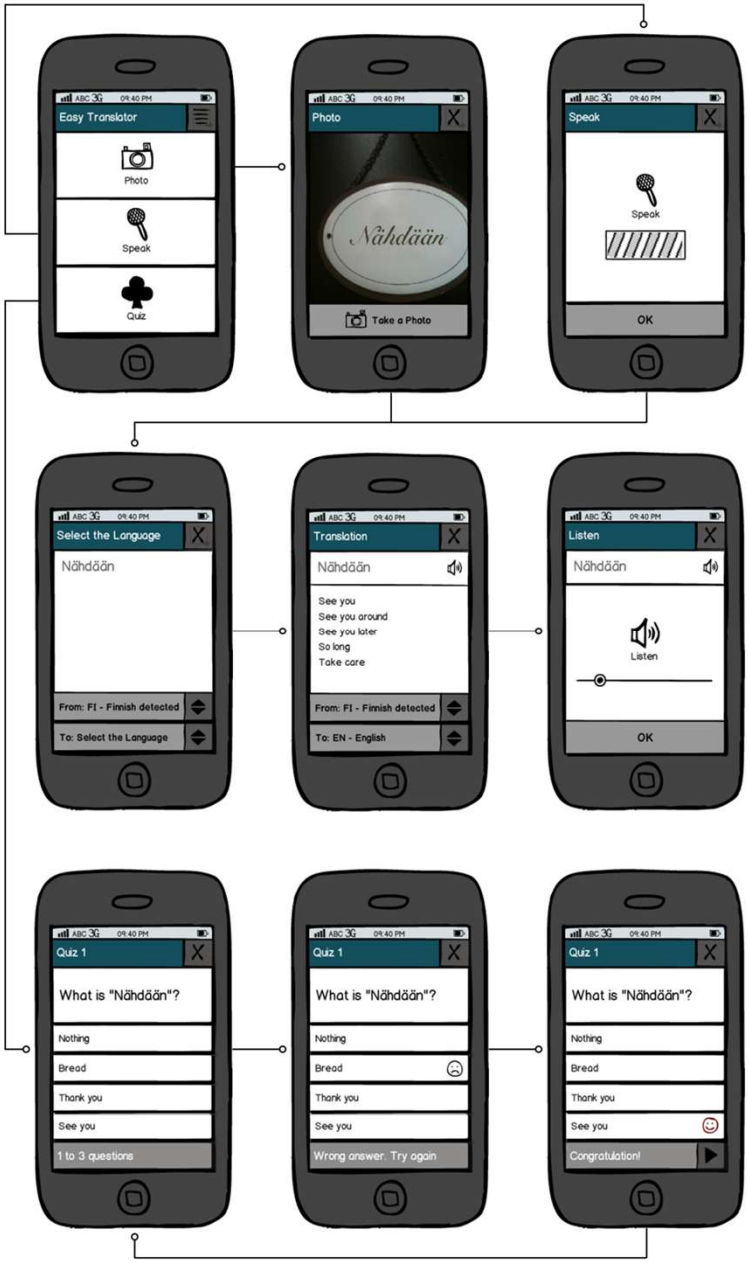
Nutrition Chart
Exercise Chart
created with Balsamiq Mockups -

ANOTHER EXAMPLE



PRELUDE: MOCKUP

- A storyboard can be ‘executed’ using a **mockup** , which is a high-fidelity static design for representing the final app

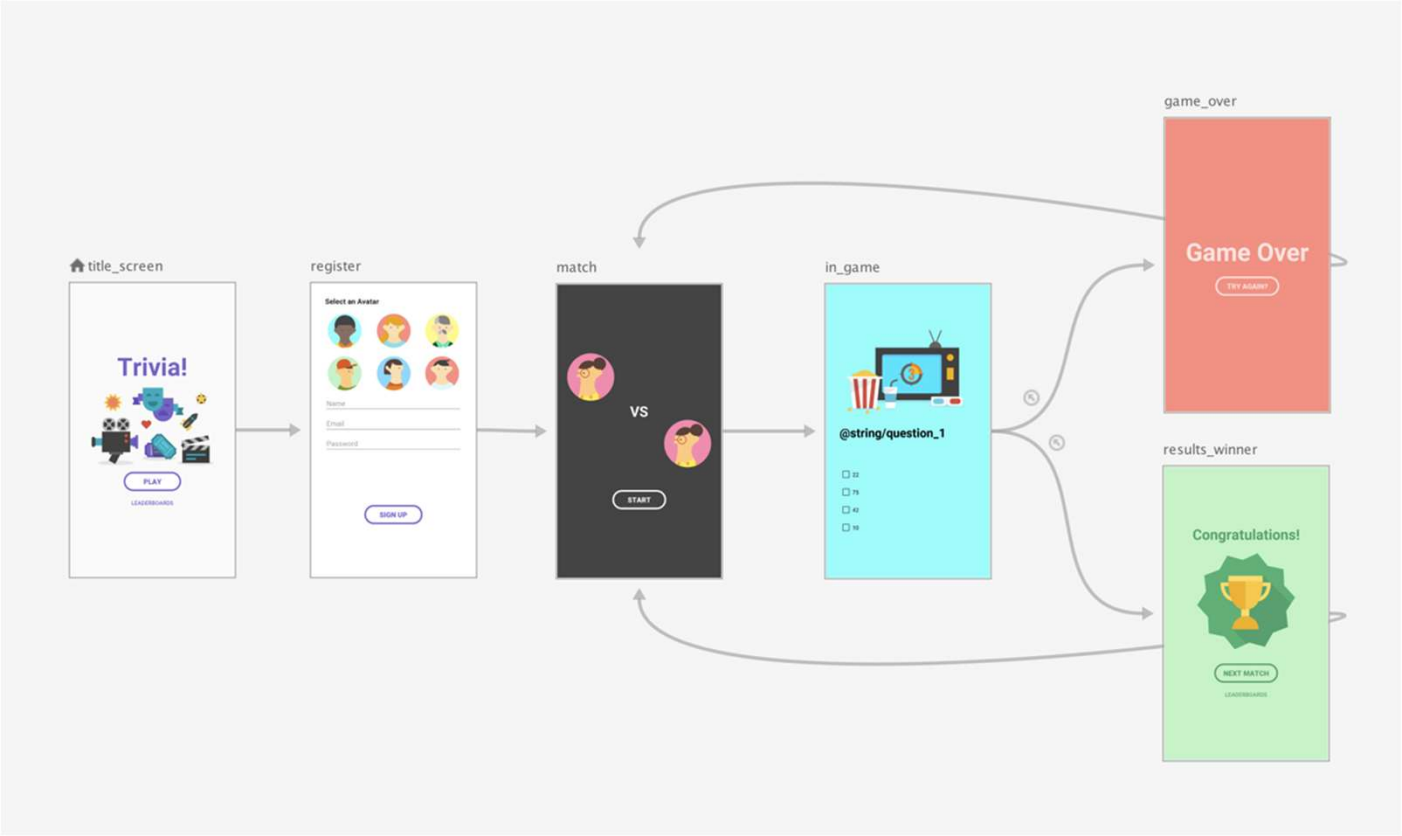
A USEFUL TOOL: FIGMA

- Figma is a collaborative, cloud-based interface design tool used for creating websites, apps, and other digital products.

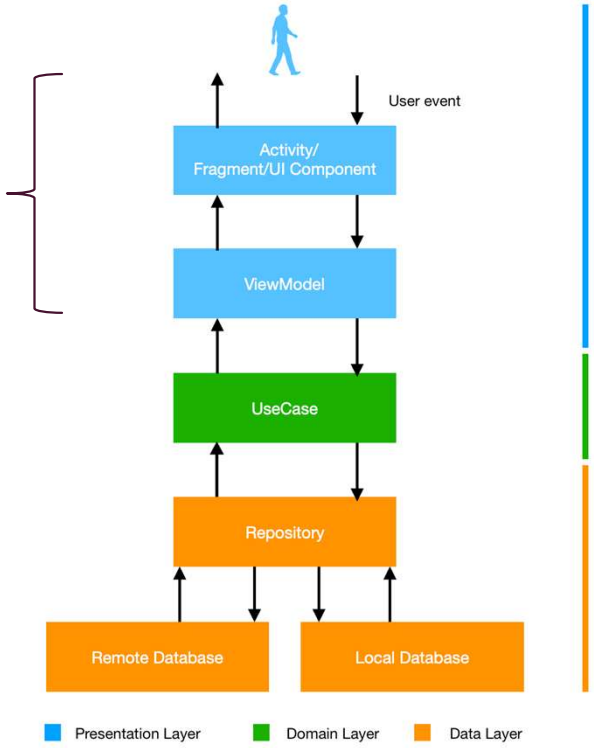
https://www.figma.com/design/IlsGy55CL4NAMG3A5NMyu0/Prototyping-in-Figma?node-id=0-1&p=f&t=AxVRgBp16Trmngpv-0

NAVIGATION GRAPH IN ADROID

- In the Android framework, the storyboard is connected to the concept of **navigation graph**
- The navigation graph is a graph where nodes are screens and edges actions that allows to reach a screen



CLEAN ARCHITECTURE



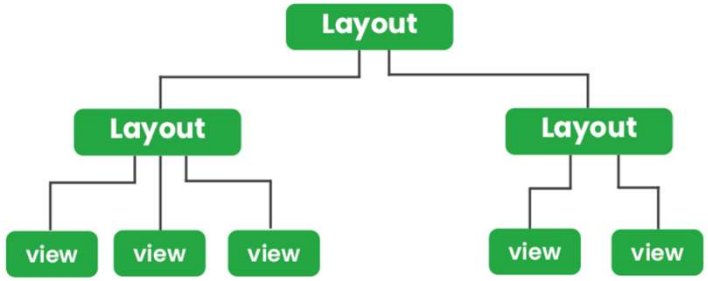
MAIN ADVANTGES

- Easy Modification
- Easy test
- Easy to reuse

THE ANDROID VIEW SYSTEM

- The area of the screen that can draw UI elements is considered as a **tree of views**
- **View** = rectangular shape on the screen that 'knows how to draw itself' wrt to the containing view
- View can be populated using

1. UI toolkits in the Android framework are based on:
   - Imperative language (the 'old' way) using XML files describing UI building blocks, like HTML
   - Declarative language using composable functions in androidx.
2. Draw directly via Canvas abstractions (2D graphics or openGL ES)



JETPACK'S COMPOSABLE FUNCTIONS

- A composable function takes data as input and emits UI elements

```
@Composable
fun Greeting(name: String) {
    Text("Hello $name")
}
```

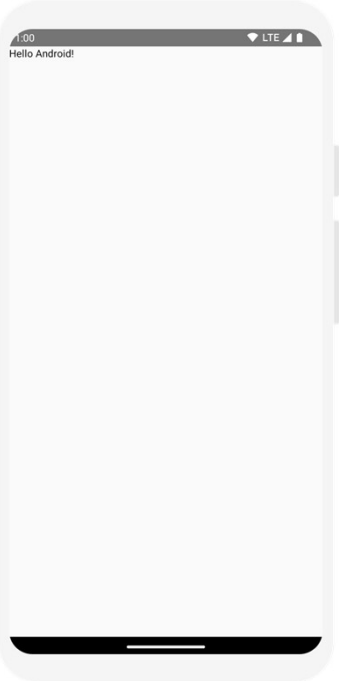
COMPOSE. FUNDAMENTALS

- Jetpack Compose is built around composable functions.
- UI is described programmatically by composable functions describing how UI should look
- A composable function has the annotation @Composable on top of the function name.

```
class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContent {
         MessageCard("Android")
      }
   }
}

@Composable
fun MessageCard(name: String) {
   Text(text = "Hello $name!")
}
```

@Preview annotation shows the function in AS



COMPOSABLE FUNCTIONS

- Composable functions can be composed, meaning that they can combine other functions in their body by calling the functions
- In this way, UI is a hierarchy of composable function
- For example, a screen might be composed of a Column containing Text, Image, and Button composables.
- The leave of this hierarchy are basic UI elements (such as Text, Button, etc.)

```
class MainActivity : ComponentActivity() {
  override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContent {
      MessageCard(Message("Android", "Jetpack Compose"))
    }
  }
}

data class Message(val author: String, val body: String)

@Composable
fun MessageCard(msg: Message) {
  Text(text = msg.author)
  Text(text = msg.body)
}

@Preview
@Composable
fun PreviewMessageCard() {
  MessageCard(
    msg = Message("Lexi", "Hey, Compose, it's great!")
  )
}
```

LAYOUT

- Some function are containers of others
- The Column function allows to arrange elements vertically.
- The Row function arranges items horizontally and Box to stacks elements

```
fun MessageCard(msg: Message) {
    Column {
        Text(text = msg.author)
        Text(text = msg.body)
    }
}
```

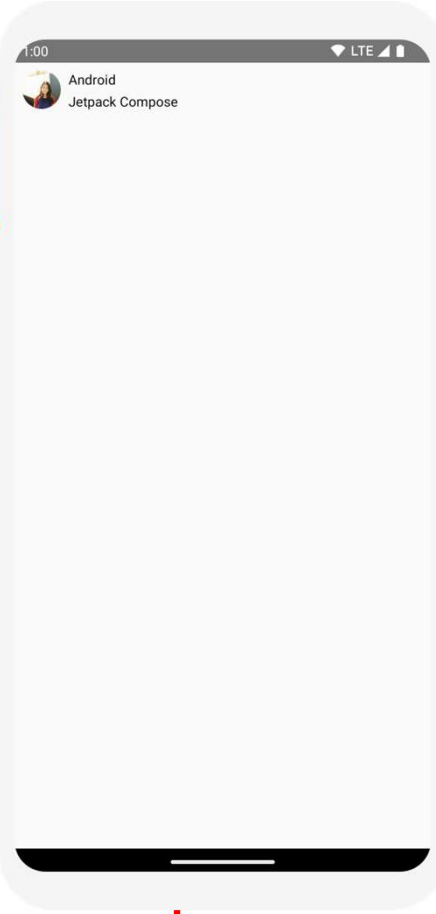
MODIFIERS

- To decorate or configure a composable, Compose uses **modifiers**.
- They allow to change the composable's size, layout, appearance or add high-level interactions, such as making an element clickable.

```
@Composable
fun MessageCard(msg: Message) {
  // Add padding around our message
  Row(modifier = Modifier.padding(all = 8.dp)) {
    Image(
      painter = painterResource(R.drawable.profile_picture),
      contentDescription = "Contact profile picture",
      modifier = Modifier
        // Set image size to 40 dp
        .size(40.dp)
        // Clip image to be shaped as a circle
        .clip(CircleShape)
    )

    // Add a horizontal space between the image and the column
    Spacer(modifier = Modifier.width(8.dp))

    Column {
      Text(text = msg.author)
      // Add a vertical space between the author and message texts
      Spacer(modifier = Modifier.height(4.dp))
      Text(text = msg.body)
    }
  }
}
```



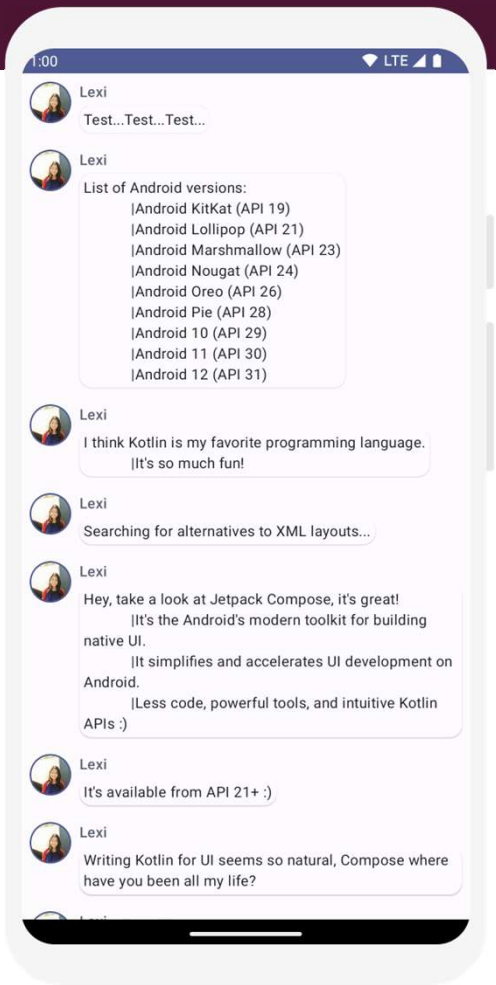
EXAMPLE: LIST

```
import androidx.compose.foundation.lazy.LazyColumn

import androidx.compose.foundation.lazy.items

    @Composable
    fun Conversation(messages: List<Message>) {
        LazyColumn {
            items(messages) { message ->
                MessageCard(message)
            }
        }
    }

    @Preview
    @Composable
    fun PreviewConversation() {
        ComposeTutorialTheme {
            Conversation(SampleData.conversationSample)
        }
    }
```



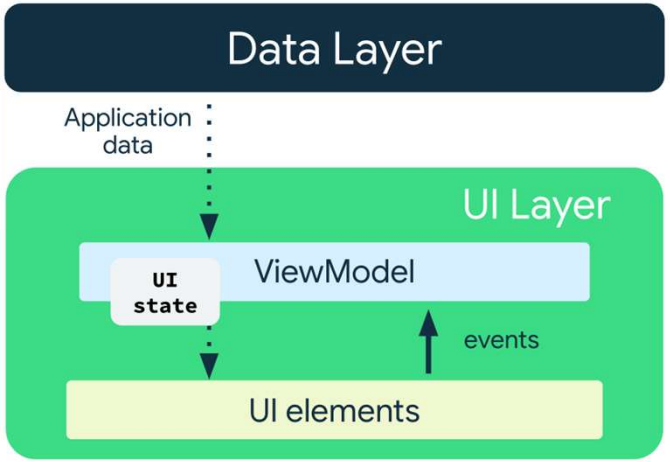
STYLES AND MATERIAL DESIGN

- Compose is built to support Material Design principles. Many of its UI elements implement Material Design out of the box.
- Material Design is built around three pillars: Color, Typography, and Shape.
- Material design style is applied using a Theme
- Scaffold provides the high-level layout structure for a typical Material Design screen.
- For example, it gives "slots" for:
  - topBar: A bar at the top (e.g. TopAppBar).
  - bottomBar: A navigation bar at the bottom.
  - floatingActionButton: The floating action button.
  - drawerContent: The drop-down side menu.

```
class MainActivity : ComponentActivity() {
  override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    enableEdgeToEdge()
    setContent {
      MyApplicationTheme {
        Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
          Greeting(
            name = "Android",
            modifier = Modifier.padding(innerPadding)
          )

      }
    }
  }
}
```

VIEW AND VIEWMODEL



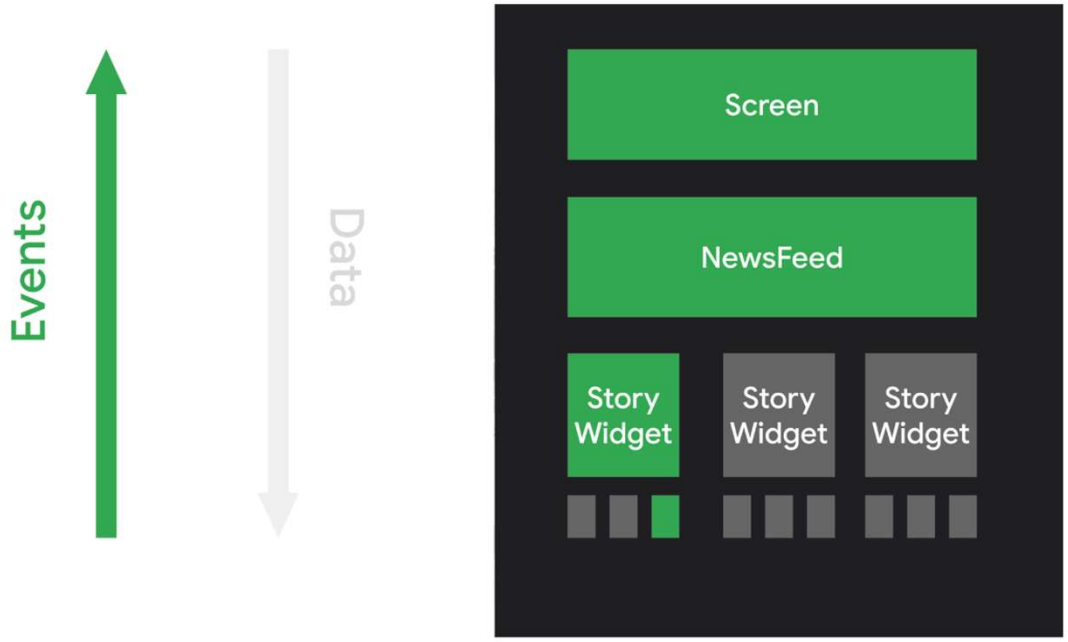
- Recommended design principle:

1. **The UI elements should be stateless**
2. **Dependency Injection: provide dependencies (necessary objects) as argument, instead of creating them directly within it.**
3. **ViewModel holds only the state of the UI elements**

UDF is obtained using two key data types
MutableStateFlow<T> → container of values that can change (state=has a value, flow=emits notification)
StateFlow<T> → Read only version of a mutable (not mutable)

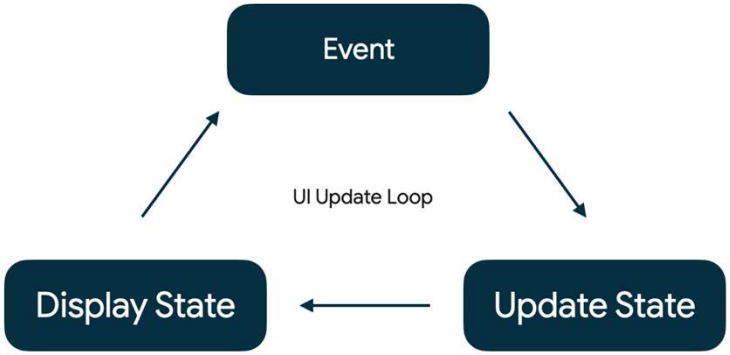
COMPOSABLE FUNCTIONS

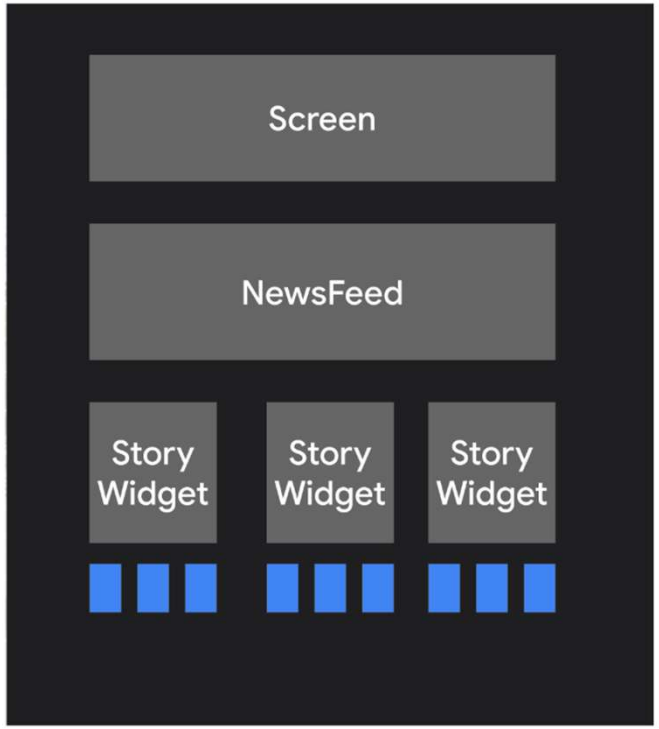
- When the user interacts with the UI, UI raises events such as onClick.
- Those events should notify the app logic, which can then change the app's state.
- Note: In this figure the data is on top



UDF

- Data flow in one direction (in this figure
- When the state changes, the composable functions are called again with the new data.
- This causes the UI elements to be redrawn--this process is called ***recomposition***.





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- Resources

Theming

Android Developers > Modern Android > Compose > Guides

Was this helpful?

Button

Buttons are fundamental components that allow the user to trigger a defined action. There are five types of buttons. The following table describes the appearance of each of the five button types, as well as where you should use them.

Type	Appearance	Purpose
Filled	Solid background with contrasting text.	High-emphasis buttons. These are for primary actions in an application, such as "submit" and "save." The shadow effect emphasizes the button's importance.
Filled tonal	Background color varies to match the surface.	Also for primary or significant actions. Filled buttons provide more visual weight and suit functions such as "add to cart" and "Sign in."
Elevated	Stands out by having a shadow.	Fits a similar role to tonal buttons. Increase elevation to cause the button to appear even more prominently.
Outlined	Features a border with no	Medium-emphasis buttons, containing actions that are important but not

On this page

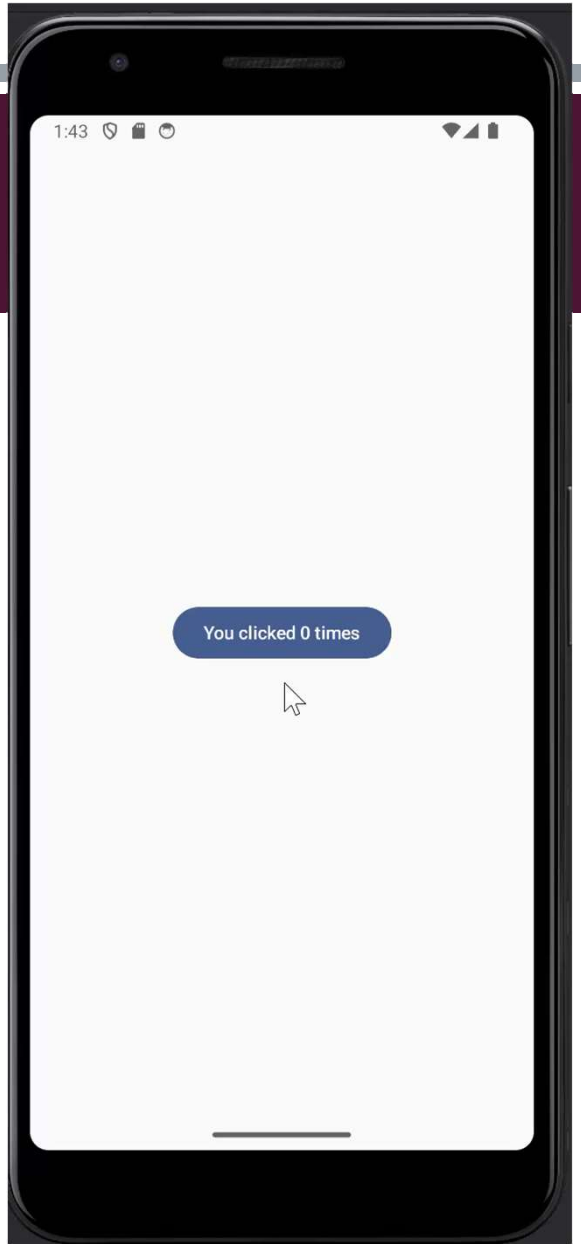
- API surface
- Filled button
- Filled tonal button
- Outlined button
- Elevated button
- Text button
- Additional resources

CODE GENERATION VIA PROMPT

- The code produced by the agent, depends on how one details the requests, paraphs with some example, so do drive the reply
- This implies to know the architectural principal of the language
- For example, a prompt like "*write code for a stateful button*" will not generate any viewmodel

CODING

- Write code with a stateful button



BASIC EXAMPLE (STATEFUL BUTTON)

- 1 ViewModel has a private and editable status: _count: MutableStateFlow.
- 2. ViewModel exposes a public and unmodifiable state to the outside world: count: StateFlow.
- 3. UI (CounterScreen) uses .collectAsState() To read and listen to count.
- 4. The user clicks the Button.
- 5. The event counterViewModel.incrementCount() is called .
- 6. ViewModel uses .update { ... } To securely change your private state _count.
- 7. The change of_count is automatically issued by count.
- 8. collectAsState() in the UI receives the new value and forces a recomposition.
- 9. UI updates to show the new number.